Task 2: Wazuh Integration with Pfsense and Firewall Rules

Instructor: Muhammad Zain Ashraf

Name: Muhammad Abdullah

Email Address: muhammadabdullahsalahuddin@gmail.com

Date: 29/07/2025

1. Introduction

The objective of this task was to design, implement, and validate a controlled and monitored virtual network environment in which a **software-based firewall** operates as the central security component, integrated with a **Security Information and Event Management (SIEM)** platform (Wazuh).

The work involved:

- Deploying **pfSense** as a software-based firewall inside a VMware environment.
- Configuring firewall rules to block traffic based on **geolocation**, **specific domains**, and **administrative access privileges**.
- Integrating pfSense with **Wazuh** for real-time security monitoring using Syslog.
- Testing and validating the rules through a simulated lab environment.
- Conducting **traffic analysis** based on the monitored logs.

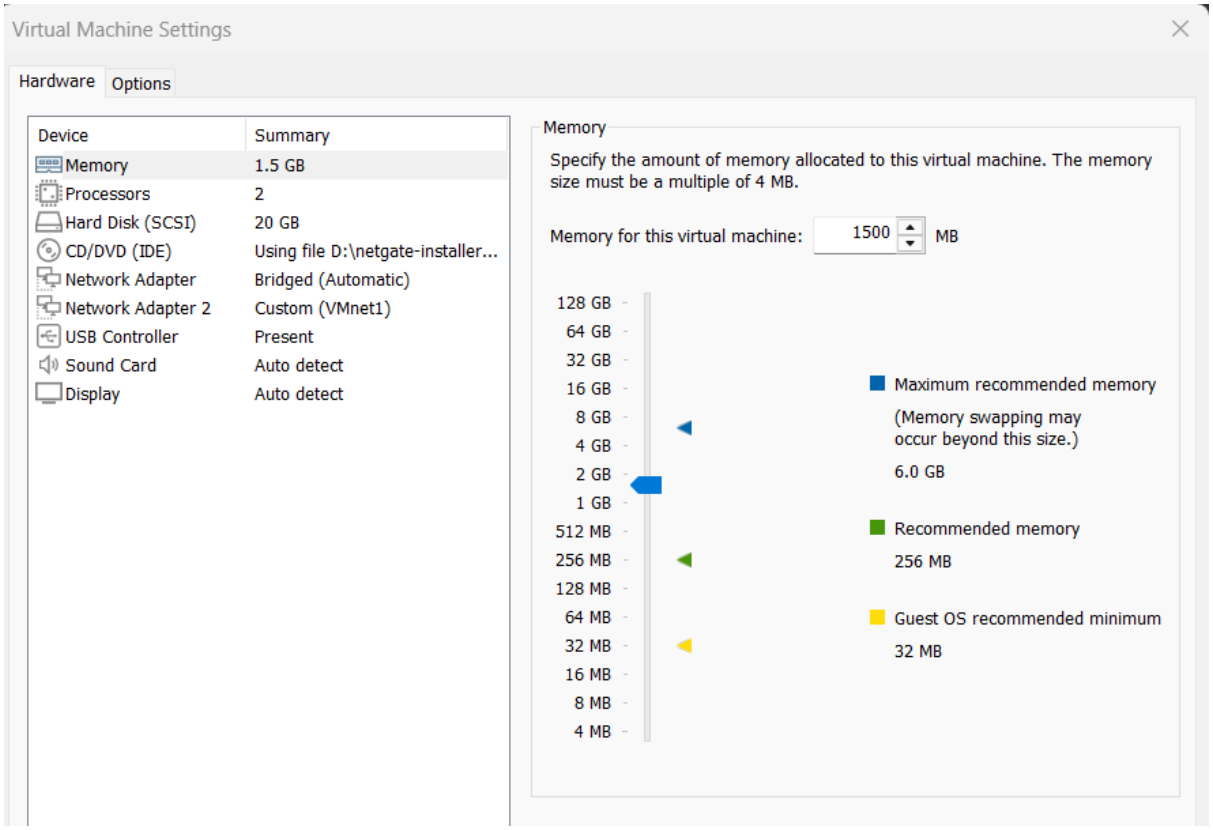
2.Environment Setup & Network Topology

2.1 Virtual Machines Used

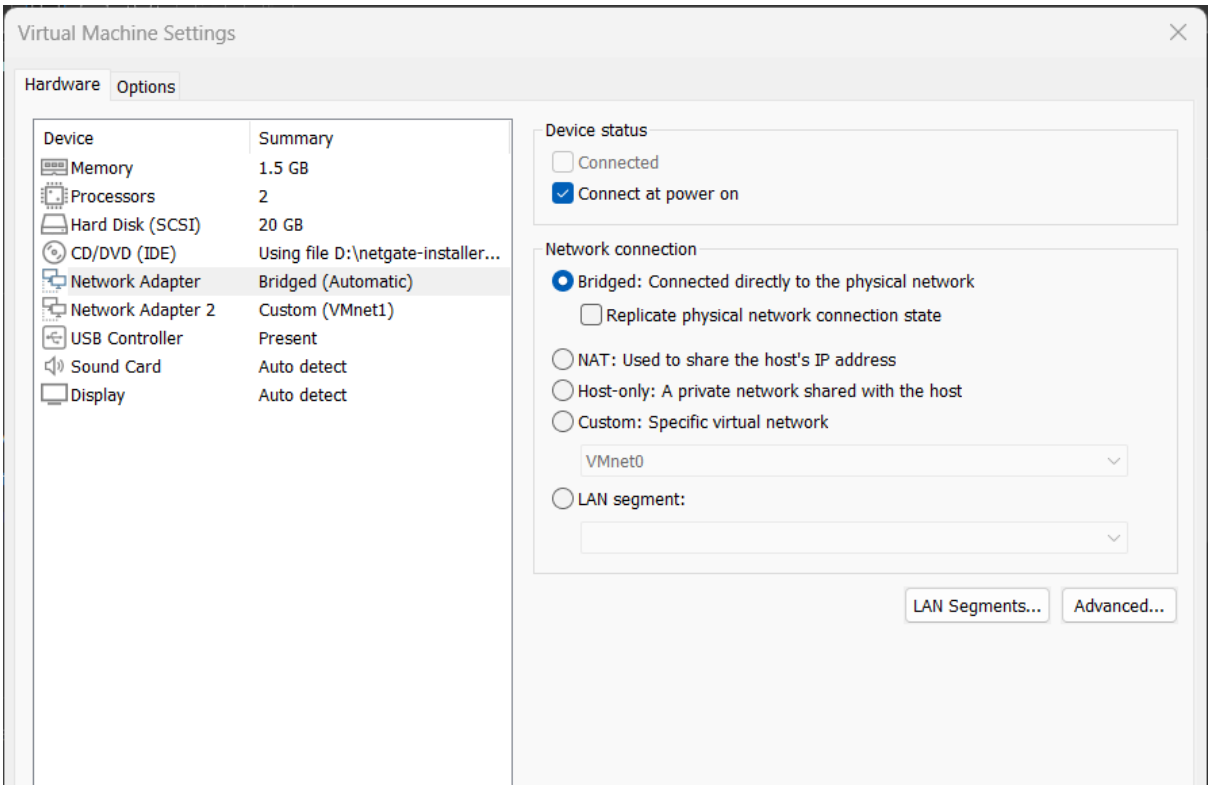
1. **Ubuntu 22.04** – Multi-role system:
    - **Role 1:** Wazuh Manager host for SIEM operations.
    - **Role 2:** Internal workstation routing all traffic through pfSense for inspection.
    - **IP Address:** 192.168.213.130 (VMnet1 – Host Only network).
2. **pfSense Firewall** – Network security and traffic control:
    - **LAN Interface:** 192.168.213.1 (VMnet1 – Host Only network).
    - **WAN Interface:** 192.168.18.130 (Bridged network to external internet).
    - **WAN Role:** Acts as the gateway between the internal VMnet1 network and the external network (internet).

2.2 Network Design

- **VMware Network Configuration:**
    - **VMnet1 (Host Only):** Internal, isolated network connecting Ubuntu and pfSense LAN.



- **Bridged Network:** pfSense WAN directly connected to the host machine's physical network adapter, allowing internet access.



- **Routing:**
    - Ubuntu uses pfSense LAN IP (192.168.213.1) as its **default gateway**.
    - All outbound internet-bound packets from Ubuntu pass through pfSense's LAN → WAN route, ensuring inspection and rule enforcement.

Diagram – Logical Topology:

Ubuntu 22.04 / Wazuh Manager

192.168.213.130

| |

| |

VMnet1 (Host Only)

| |

| |

LAN - pfSense Firewall

192.168.213.1

| |

| |

WAN - pfSense Firewall

192.168.18.130

| |

| |

Internet (Bridged)

---

3. Firewall Implementation

3.1 Software-Based Firewall Choice – pfSense

pfSense was chosen because it is a robust, open-source firewall platform supporting:

- **Stateful packet inspection (SPI)**
- **Advanced routing features**

- **GeoIP filtering via pfBlockerNG**
- **Syslog integration for SIEM platforms**
- **Granular access control rules**

3.2 pfBlockerNG Setup & MaxMind GeoIP Integration

- Installed **pfBlockerNG-devel** package to enable advanced threat filtering and GeoIP blocking.
- Created a **MaxMind GeoLite2 Lite** account and acquired the license key.
- Integrated the MaxMind database with pfBlockerNG to map IPs to geolocations.
- Created and activated blocklists for **Russia** and **China**, enabling both **inbound** and **outbound** blocking policies.

---

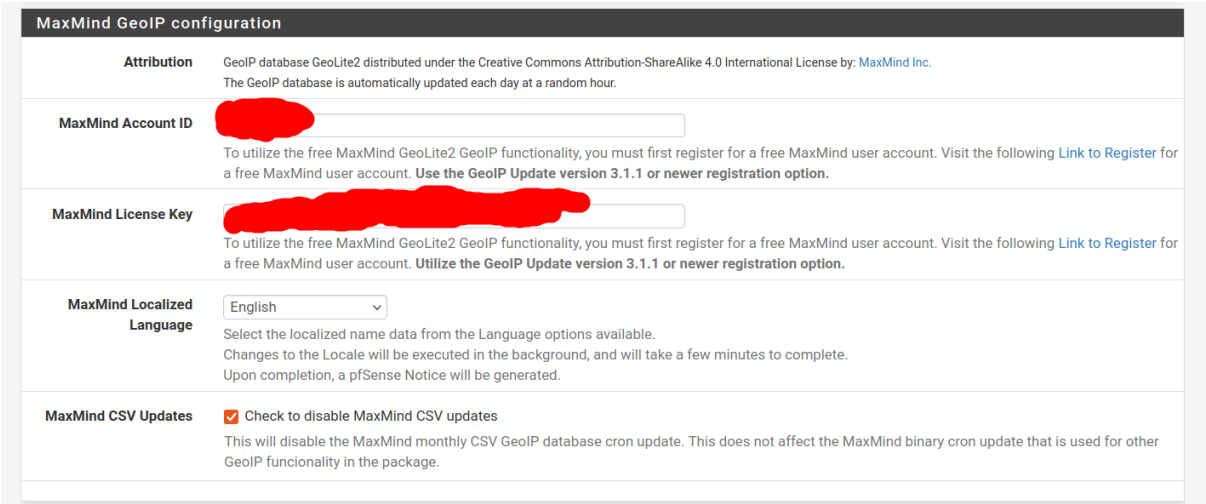
4. Firewall Rules Configuration & Testing

4.1 Geolocation Blocking (Russia & China)

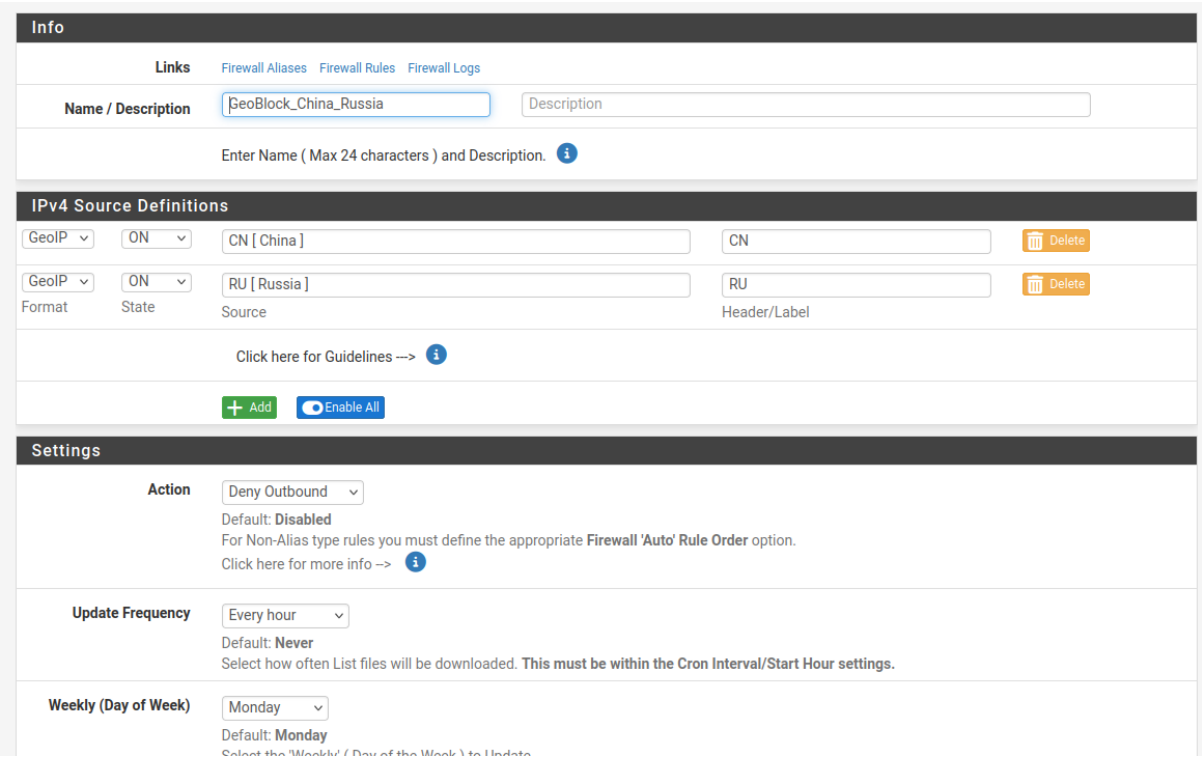
Objective: Mitigate risks from high-threat regions by blocking all traffic associated with Russian and Chinese IP ranges.

Configuration Steps:

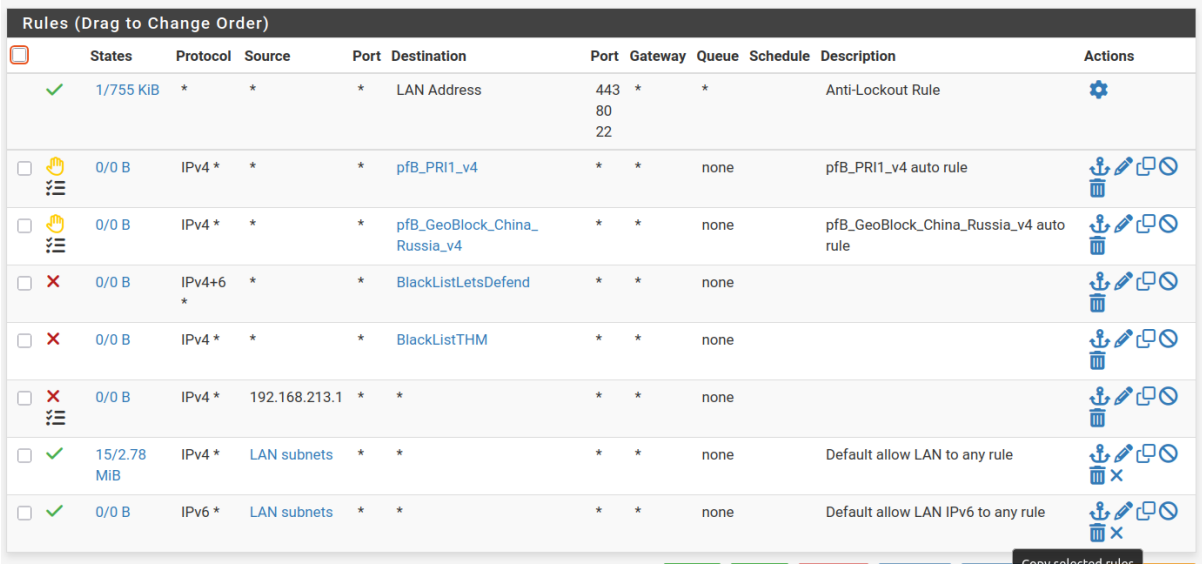
1. Enabled **GeoIP** under pfBlockerNG.



2. Created a GeoIP alias containing **RU (Russia)** and **CN (China)**.



3. Applied the alias to firewall rules on both **WAN** and **LAN** interfaces:
    - **Outbound Rule:** Prevent internal hosts from connecting to blocked regions.
    - **Inbound Rule:** Block external connections originating from blocked regions.



Testing:

- Attempted to browse websites hosted in the blocked regions.
- Verified blocked connections via pfSense firewall logs.

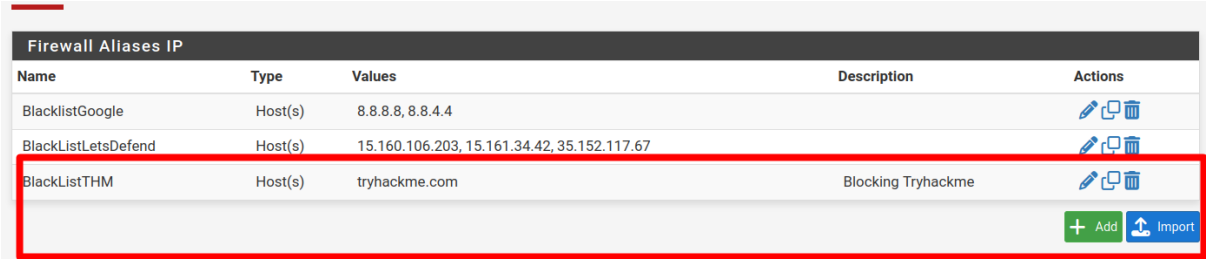
- Confirmed corresponding **Security Event entries** in Wazuh showing “GeoIP Block” actions.

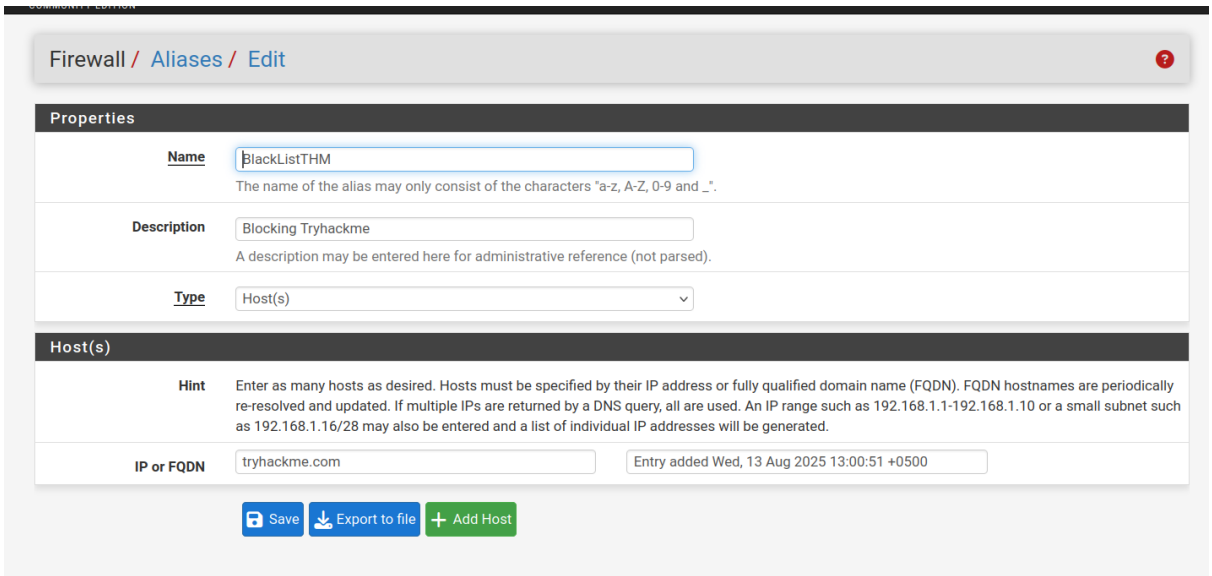
4.2 Website Blocking (tryhackme.com)

Objective: Demonstrate domain-specific access restrictions.

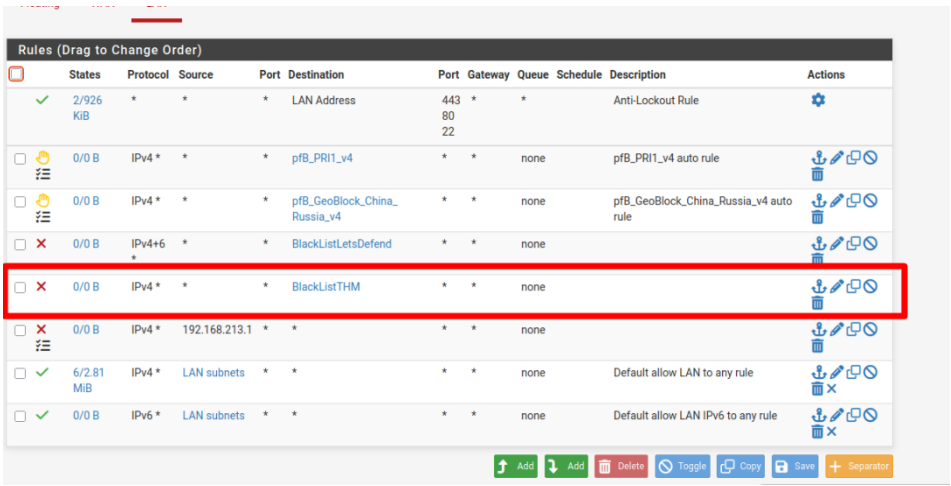
Configuration Steps:

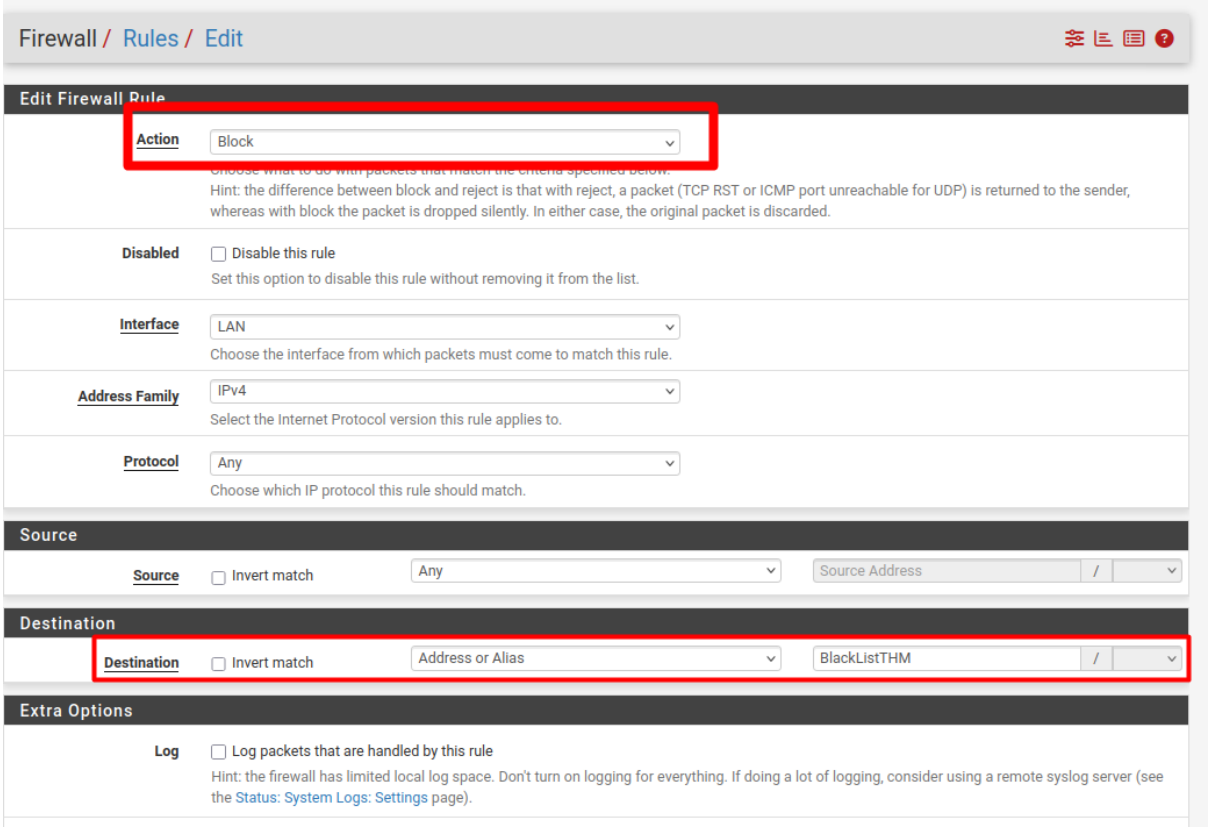
1. Created a firewall alias for tryhackme.com.





2. Configured a LAN rule to **block DNS resolution and HTTP/HTTPS requests** to that domain.





3. Added logging for the rule to monitor access attempts.

Testing:

- Browsed tryhackme.com from Ubuntu – access denied.
- Wazuh logs captured the denial event with full packet details.

---

6. Wazuh SIEM Integration

6.1 Integration Method

- **Syslog Output** from pfSense was directed to Ubuntu's Wazuh Manager.
- Wazuh was configured to parse incoming pfSense log entries using its firewall decoder rules.

timestamp per 30 minutes

Time	agent.name	rule.description	rule.level	rule.id
Aug 15, 2025 @ 10:05:46.578	thebe562-virtual-machine	Multiple pfSense firewall blocks events from same source.	10	87702
Aug 15, 2025 @ 10:00:46.790	thebe562-virtual-machine	Multiple pfSense firewall blocks events from same source.	10	87702
Aug 15, 2025 @ 09:56:33.873	thebe562-virtual-machine	PAM: Login session opened.	3	5501
Aug 15, 2025 @ 09:56:33.872	thebe562-virtual-machine	PAM: Login session opened.	3	5501
Aug 15, 2025 @ 09:55:19.801	thebe562-virtual-machine	PAM: Login session opened.	3	5501
Aug 15, 2025 @ 09:55:19.795	thebe562-virtual-machine	PAM: Login session opened.	3	5501
Aug 15, 2025 @ 09:55:11.508	thebe562-virtual-machine	syslog: User authentication failure.	5	2501
Aug 15, 2025 @ 09:55:11.506	thebe562-virtual-machine	syslog: User authentication failure.	5	2501
Aug 15, 2025 @ 09:45:50.682	thebe562-virtual-machine	Interface entered in promiscuous(sniffing) mode.	8	5104
Aug 15, 2025 @ 09:45:50.679	thebe562-virtual-machine	Syslogd restarted.	5	1005
Aug 15, 2025 @ 09:45:50.678	thebe562-virtual-machine	Interface entered in promiscuous(sniffing) mode.	8	5104
Aug 15, 2025 @ 09:45:50.666	thebe562-virtual-machine	Interface entered in promiscuous(sniffing) mode.	8	5104

-

.

7. Traffic Analysis

- **Geolocation Filtering:** Effectively blocked multiple unsolicited connection attempts from blacklisted countries. These appeared predominantly as TCP SYN packets attempting to initiate connections to internal services.
- **Website Blocking:** Requests to tryhackme.com were dropped before DNS resolution could occur, eliminating the possibility of circumvention

---

8. Security Benefits

1. **Reduced Attack Surface:** Geo-blocking eliminated traffic from regions with high malicious activity.
2. **Policy Enforcement:** Website restrictions ensured compliance with operational policies.
3. **Centralized Monitoring:** Wazuh provided consolidated visibility into network security events.

---

9. Conclusion

This lab successfully demonstrated:

- The deployment of **pfSense** in a controlled VMware environment.
- The implementation of **geolocation-based, domain-based, and privilege-based firewall rules**.
- The integration of pfSense with **Wazuh SIEM** for log collection, monitoring, and alerting.
- Effective blocking and monitoring of high-risk traffic.

This setup follows a **defense-in-depth** strategy, combining **network-level filtering** with **centralized security event management**, offering both proactive and reactive security capabilities suitable for SOC environments.